

Quantitative obstructions for nearly shortest Egyptian fractions

A partial-result note for Erdős problem 295

tienxion • Revised 5 September 2026

Abstract

For distinct integers at least N whose reciprocals sum to one, write D for the excess length over $(e - 1)N$. For every fixed $C \geq 0$ and $\varepsilon > 0$, we show that the number of starts $N \leq X$ admitting $D \leq C$, with unrestricted denominators, is $O_{C,\varepsilon}(X^{17/30+\varepsilon})$. We also give an exponential-denominator obstruction, explicit lower bounds under polynomial ceilings, and a matching asymptotic with coefficient $1.79033944\dots$ under the ceiling $(e^2/6)N^2$. At the smaller exact ceiling N^2 , we determine the order $N/\log N$ with different lower and upper constants. The density estimate permits infinitely many sparse exceptions; the unrestricted pointwise problem remains unresolved. The methods have classical antecedents; novelty is not asserted.

Revision status. This version adds the density theorem and the quadratic upper constructions, with their proofs. The new results have undergone informal local checks, not independent peer review or formal verification. The review of the preceding version does not certify these additions. Throughout, A is a finite set of distinct integers at least N , and

$$\sum_{a \in A} \frac{1}{a} = 1, \quad k = |A|, \quad D = k - (e - 1)N, \quad M = \max A, \quad b = \frac{e - 1}{2}.$$

All logarithms are natural. Asymptotic assertions concern sequences with $N \rightarrow \infty$. The original problem asks whether the *unrestricted* minimum possible D tends to infinity.

Results

Theorems 1–3 retain the preceding version’s lower bounds. The principal additions are Theorem 4, which allows unrestricted denominators, and Theorem 5, which supplies upper constructions under specified ceilings.

Theorem 1 (exponential tail). If $D = o(N/\log N)$, then

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \log \prod_{\substack{a \in A \\ a > N^2}} a \geq e, \tag{1}$$

and

$$\liminf_{N \rightarrow \infty} \frac{(D + b) \log M}{N} \geq e. \tag{2}$$

More precisely, with $r = |A \cap (N^2, \infty)|$,

$$1 \leq r \leq D + b + O\left(\frac{(D + 1)^2}{N}\right) \tag{3}$$

for sufficiently large N ; the lower inequality follows from (1).

Corollary 1.1 (bounded excess). Fix $C \geq 0$, and put

$$q = \left\lfloor C + \frac{e-1}{2} \right\rfloor.$$

If $q = 0$, representations with $D \leq C$ do not exist for sufficiently large N . If $q \geq 1$, every sequence of such representations satisfies

$$\liminf \frac{\log M}{N} \geq \frac{e}{q}. \quad (4)$$

Thus a fixed additive excess forces a bounded number of denominators above N^2 , whose product has logarithm at least $(e - o(1))N$.

Corollary 1.2 (a subexponential ceiling). If $F = F(N)$ satisfies

$$\log N = o(\log F), \quad \log F = o(N),$$

then every representation with $M \leq F$ satisfies

$$D \geq (e - o(1)) \frac{N}{\log F}. \quad (5)$$

For example, $M \leq \exp(N^\alpha)$, $0 < \alpha < 1$, implies

$$D \geq (e - o(1))N^{1-\alpha}.$$

Theorem 2 (a polynomial ceiling). Fix $B > 1$. Every sequence of representations with $M \leq N^{B+o(1)}$ satisfies

$$D \geq (\mathcal{C}(B-1) - o(1)) \frac{N}{\log N}, \quad (6)$$

where, for real $h > 0$,

$$\mathcal{C}(h) = e \int_0^e \min \left\{ w(x), \frac{1}{eh} \right\} dx, \quad w(x) = \sum_{\substack{m \geq 1 \\ 1 \leq mx \leq e}} \left(\frac{1}{mx} - \frac{1}{e} \right).$$

Writing $c_t = \mathcal{C}(t)$ for positive integers t ,

$$\begin{aligned} c_1 &= e \log 2 - 2 + \frac{5e}{6} \log \frac{6}{5} + \frac{e}{2} \log 3 = 1.79033944 \dots, \\ c_2 &= e \log \frac{3}{2} - \frac{3}{2} + \frac{e}{2} \log 3 = 1.09533735 \dots, \\ c_t &= e \log \left(1 + \frac{1}{t} \right) \quad (t \geq 3). \end{aligned} \quad (7)$$

In particular, $M \leq N^{2+o(1)}$ gives the constant c_1 , $M \leq N^{3+o(1)}$ gives c_2 , and for $B \geq 4$ the constant is $e \log(B/(B-1))$. The component argument below is stronger than simply counting at most $\lfloor B \rfloor$ large primes per denominator.

Theorem 3 (subquadratic multiplicity). If $1 < B < 2$ and $M \leq N^{B+o(1)}$, put $s = \lceil 1/(B-1) \rceil$. Then

$$D \geq (d_s - o(1)) \frac{N}{\log N}, \quad d_s = \mathcal{C}(1/s) = e \int_0^e \min \{ w(x), s/e \} dx. \quad (7a)$$

More generally, for each fixed integer $s \geq 2$, the same bound holds if

$$M = o\left(N^{1+1/(s-1)}\right). \quad (7b)$$

Thus $M = o(N^2)$ yields $d_2 = 2.59516741\dots$, whereas Theorem 2 supplies $c_1 = 1.79033944\dots$ at $N^{2+o(1)}$. Also $d_s = \log s + O(1)$ as $s \rightarrow \infty$. These are lower-bound constants, not claimed optimal constants.

Theorem 4 (unrestricted density bound). Let $k(N)$ denote the minimum length with no upper restriction on the denominators, and set

$$\mathcal{E}_C = \{N \geq 2 : k(N) - (e-1)N \leq C\}.$$

For every fixed $C \geq 0$ and $\varepsilon > 0$,

$$\#(\mathcal{E}_C \cap [2, X]) \ll_{C,\varepsilon} X^{17/30+\varepsilon}. \quad (\text{D})$$

More precisely, if $q = \lfloor C+b \rfloor \geq 1$, then for every fixed $17/30 < \theta < 0.99$ the bound is $O_{C,\theta}(X^\theta (\log X)^{2q})$. If $q = 0$, the set is finite. The exponent uses Guth and Maynard's theorem on primes in short intervals [8, Corollary 1.3]. All constants may depend on the fixed excess bound; no assertion for growing C follows by substituting $C = C(X)$.

Theorem 5 (quadratic upper constructions). For fixed $\lambda > 0$, let $k_\lambda(N)$ be the minimum length when every denominator lies in $[N, \lambda N^2]$, with value $+\infty$ if no representation exists. Put $\kappa = e^2/6$. For all sufficiently large N , these minima are finite for $\lambda = 1, \kappa$, and

$$k_\kappa(N) - (e-1)N = (c_1 + o(1)) \frac{N}{\log N}. \quad (\text{Q})$$

At the exact ceiling N^2 , the bounds are

$$(c_1 - o(1)) \frac{N}{\log N} \leq k_1(N) - (e-1)N \leq \left(4 - \frac{e}{2} \log 2 + o(1)\right) \frac{N}{\log N}, \quad (\text{Q1})$$

where $4 - (e/2) \log 2 = 3.057915307318\dots$. Thus the order is $N/\log N$ for every fixed ceiling N^B , $B \geq 2$. The matching constant in (Q) is proved for $(e^2/6)N^2$, not for N^2 .

Classical comparison and provenance. The question comes from Erdős and Graham [1, p. 35]. Vose's classical theorem [4], as restated in [5, p. 151, equation (2)], implies

$$k(N) \leq (e-1)N + O(\sqrt{N}).$$

Indeed, take the consecutive block immediately before its reciprocal sum first reaches one. Its positive residual is at most the reciprocal of the next denominator, and has reduced denominator with logarithm $O(N)$. Vose expands the residual into $O(\sqrt{N})$ distinct unit fractions. Each new denominator is at least the next denominator, so these terms are disjoint from the block. This is an attributed classical corollary.

The prime-divisor and positive-numerator methods below have classical antecedents, notably Croot [2] and Martin [3]. The upper constructions use their two-fraction cancellation device, with an elementary proof of the form needed here. The almost-everywhere conclusion combines a local comparison argument with the established prime-distribution theorem [8]; that analytic input is not a result of this note. The matching construction also uses the classical upper sieve in [9].

The original 1971 solution [6] has now been read. It obtains the historical $O(N/\log N)$ unrestricted upper excess by applying an earlier Egyptian-fraction length theorem to the residual of a consecutive block. It does not establish the simultaneous quadratic ceilings used here. Burshtein's relevant 2005 paper [7] remains uninspected in full. The source search and informal checks do not establish novelty or priority for the formulations in this note.

1. The harmonic loss identity

Set

$$L = N + k - 1 = eN + D - 1, \quad I = [N, L] \cap \mathbb{Z}, \quad H = I \setminus A, \quad E = A \setminus I.$$

Since $|H| = |E|$,

$$\delta := \sum_{j=N}^L \frac{1}{j} - 1 = \sum_{h \in H} \left(\frac{1}{h} - \frac{1}{L} \right) + \sum_{a \in E} \left(\frac{1}{L} - \frac{1}{a} \right) \geq 0. \quad (8)$$

The nonnegativity also follows by ordering the elements of A .

The elementary bound

$$1 \leq \frac{1}{N} + \log \frac{L}{N}$$

gives $L \geq Ne^{1-1/N}$, hence D is bounded below by an absolute constant. When $D = o(N)$, the expansion

$$H_n = \log n + \gamma + \frac{1}{2n} + O(n^{-2})$$

gives

$$eN\delta = D + b + O\left(\frac{(|D| + 1)^2}{N}\right).$$

When $D < 0$, the already established absolute lower bound makes the error $O(1/N)$, and $\delta \geq 0$ yields $D \geq -b - O(1/N)$. When $D \geq 0$, no lower estimate is needed. Consequently $D + 1$ is bounded below by a positive constant for large N , and $|D| + 1 \ll D + 1$. We may therefore use the form

$$eN\delta = D + b + O\left(\frac{(D + 1)^2}{N}\right). \quad (9)$$

In particular, for $D = o(N/\log N)$ or $D = O(N/\log N)$, one has $\delta = O((D + 1)/N)$.

2. A bounded-cofactor prime lemma

Lemma. Fix positive ε, T . For sufficiently large N , if a prime $p > \varepsilon N$ divides some element of $A \cap [N, TN]$, then p divides some element of $A \cap (TN, \infty)$.

Proof. Suppose there is no such latter element. Write the selected multiples of p as mp , with m belonging to a nonempty subset $J \subseteq \{1, \dots, K\}$, where $K = \lfloor T/\varepsilon \rfloor + 1$ is fixed. Their reciprocal sum is

$$\frac{1}{p} \sum_{m \in J} \frac{1}{m} = \frac{u}{pv}.$$

We may take $v = \text{lcm}(1, \dots, K)$ and $1 \leq u \leq vH_K$. For sufficiently large N , $p > \max(v, vH_K)$, so the sum has p -adic valuation -1 . The reciprocals of all other elements of A , and the integer 1, have denominators coprime to p . This contradicts the equality of their difference with $u/(pv)$. $\square$

This is an elementary form of a classical prime-divisor obstruction, not a newly introduced mechanism. Croot and Martin use quantitatively stronger versions when the maximum denominator is near N ; see [2] and [3].

3. Proof of Theorem 1

Fix $0 < \varepsilon < 1$, $0 < \eta < e - 2$, and $T > e + 1$. All three parameters stay fixed until after the limit in N .

For every prime

$$\varepsilon N < p \leq (e - \eta)N,$$

choose $n_p = p \lfloor N/p \rfloor$. If $p \leq N$, then $n_p < N + p \leq 2N$; if $p > N$, then $n_p = p \leq (e - \eta)N$. Hence $n_p \in I$, a positive proportion of N below L , for sufficiently large N .

If $n_p \notin A$, its contribution to (8) is at least c_η/N , for some $c_\eta > 0$. Also, distinct primes greater than εN cannot divide the same $n_p \leq L$, since $\varepsilon^2 N^2 > L$ eventually. Equations (8)-(9) therefore show that the number of these exceptional primes is $O_\eta(D + 1)$.

For every other prime in the displayed interval, the lemma forces a multiple in $A \cap (TN, \infty)$. Thus

$$\log \prod_{\substack{a \in A \\ a > TN}} a \geq \vartheta((e - \eta)N) - \vartheta(\varepsilon N) - O_\eta((D + 1) \log N), \quad (10)$$

where $\vartheta(x) = \sum_{p \leq x} \log p$.

By (8), every denominator above TN costs at least $1/L - 1/(TN) \gg_T 1/N$. There are $O_T(D + 1)$ such denominators. Those at most N^2 contribute altogether $O_T((D + 1) \log N) = o(N)$ to the logarithm in (10). The prime number theorem now gives

$$\log \prod_{\substack{a \in A \\ a > N^2}} a \geq (e - \eta - \varepsilon - o(1))N.$$

First taking the lower limit in N , and then letting $\varepsilon, \eta \downarrow 0$, proves (1).

For $a > N^2$, the cost in (8) is at least $1/L - 1/N^2$. Therefore

$$r \leq \frac{\delta}{1/L - 1/N^2} = D + b + O\left(\frac{(D + 1)^2}{N}\right), \quad (11)$$

which proves (3). Since $r \geq 1$, this also implies $D + b \geq 1 - o(1)$ along bounded- D subsequences. In general the error in (11) is $o(D + b)$: for unbounded positive D , its relative size is $O(D/N)$; for bounded D , use the preceding lower bound.

Finally,

$$r \log M \geq \log \prod_{a > N^2} a \geq (e - o(1))N.$$

Divide using $r \leq (D + b)(1 + o(1))$. This proves (2) without multiplying an uncontrolled additive error by $\log M$.

For Corollary 1.1, (11) and integrality give $r \leq q$ eventually. Equation (1) gives (4). As a further elementary consequence, the unrestricted minimum satisfies

$$\liminf_{N \rightarrow \infty} (k(N) - (e - 1)N) \geq \frac{3 - e}{2}.$$

This is only a constant lower bound.

For Corollary 1.2, a sequence violating (5) by a fixed positive proportion would have $D = O(N/\log F) = o(N/\log N)$. Applying (2) and $\log M \leq \log F$ contradicts that violation. Here the constant b is negligible because $N/\log F \rightarrow \infty$.

4. A component bound for the number of supported primes

Fix $\varepsilon, T > 0$, with $T > e + 1$, and suppose $M \leq N^{B+\xi_N}$, where $B > 1$ is fixed and $\xi_N \rightarrow 0$. Consider all selected denominators above TN . Form a hypergraph: its vertices are their prime divisors greater than εN , and each denominator with at least one such divisor contributes an edge consisting of those distinct prime divisors. Different denominators may give the same edge; keep them as distinct edges. Each edge has at most a fixed number $J = J(B)$ of vertices for sufficiently large N .

Take one connected component with q prime vertices and $r \geq 1$ edges, represented by denominators $a_1, \dots, a_r$. Write j_i for the number of distinct large prime factors of a_i , and let b_i be the integer obtained from a_i by removing *all* prime-power factors whose primes exceed εN . Then

$$b_i \leq \frac{a_i}{(\varepsilon N)^{j_i}}. \quad (12a)$$

Add the reciprocals of the component's tail denominators and of every selected denominator at most TN divisible by one of its primes. Call this positive sum S_0 . Every complementary selected denominator is coprime to all component primes. Since the entire sum is 1, the reduced denominator of S_0 is also coprime to every component prime.

For large N , a selected denominator at most TN contains at most one prime greater than εN , to the first power. It has form mp , with $m \leq T/\varepsilon$. Set

$$L_0 = \text{lcm}(1, \dots, \lceil T/\varepsilon \rceil).$$

After the component primes cancel, the reduced denominator of S_0 therefore divides $L_0 \prod_i b_i$. Positivity implies

$$S_0 \geq \frac{1}{L_0 \prod_i b_i}. \quad (12b)$$

Each prime has at most $\lceil T/\varepsilon \rceil$ incident low denominators. Their reciprocals are at most $1/N$, while each tail reciprocal is less than $1/(TN)$. As $q \leq Jr$,

$$S_0 \leq C_{\varepsilon, T, B} \frac{r}{N}. \quad (12c)$$

This upper bound is used quantitatively; no assumption that a component has bounded size is made.

The bipartite incidence graph of a connected component has $q + r$ vertices, so

$$\sum_i j_i \geq q + r - 1.$$

Combining this fact with (12a)-(12c), and using $\log a_i \leq (B + \xi_N) \log N$, gives

$$\begin{aligned} \log N - \log(C_{\varepsilon, T, B} L_0 r) &\leq \sum_i \log b_i \\ &\leq ((B + \xi_N)r - q - r + 1) \log N + O_{\varepsilon, B}(r). \end{aligned}$$

Cancel the $\log N$ terms and rearrange:

$$q \leq (B - 1 + \xi_N)r + \frac{\log(C_{\varepsilon, T, B} L_0 r) + O_{\varepsilon, B}(r)}{\log N}. \quad (12d)$$

Sum over components. Since their number is at most the total number of nonempty edges, and $\log r \leq r$ for $r \geq 1$, all errors sum to $O_{\varepsilon, T, B}(r_T / \log N)$, where $r_T = |A \cap (TN, \infty)|$. Empty edges can only increase r_T . Thus the total number of prime vertices satisfies

$$q_{\text{total}} \leq (B - 1 + o(1))r_T. \quad (12e)$$

This is the required improvement over counting prime factors in each denominator separately. Removing all large prime *powers* in (12a) ensures that repeated factors cause no problem.

5. Proof of Theorem 2

It suffices to work along sequences with $D = O(N/\log N)$: any sequence violating (6) has this property, by the absolute lower bound for D . In this range, $L/N \rightarrow e$, and (9) gives

$$N\delta = \frac{D}{e} + o(N/\log N). \quad (12)$$

Fix $0 < \varepsilon < 1$ and $T > e + 1$. For primes $\varepsilon N < p \leq L$, define

$$W_N(p) = \sum_{\substack{m \geq 1 \\ N \leq mp \leq L}} \left(\frac{1}{mp} - \frac{1}{L} \right).$$

Partition these primes into two classes: U consists of those with no multiple in $A \cap I$, and V consists of the others.

Every core multiple of a prime in U is a hole. Core multiples corresponding to distinct such primes are disjoint for sufficiently large N , because the product of the primes exceeds L . Their total cost in (8) is consequently at least $\sum_{p \in U} W_N(p)$.

Every prime in V divides a denominator above TN , by the lemma. Put $h = B - 1$. The component bound (12e) gives $|V| \leq (h + o(1))r_T$, where $r_T = |A \cap (TN, \infty)|$. Write $h_N = h + o(1) > 0$ for a valid upper-bound coefficient. The tail's harmonic cost is at least $r_T(1/L - 1/(TN))$. Combining the two disjoint parts of (8),

$$\begin{aligned} \delta &\geq \sum_{p \in U} W_N(p) + \frac{|V|}{h_N} \left(\frac{1}{L} - \frac{1}{TN} \right) \\ &\geq \sum_{\varepsilon N < p \leq L} \min \left\{ W_N(p), \frac{1}{h_N} \left(\frac{1}{L} - \frac{1}{TN} \right) \right\}. \end{aligned} \quad (13)$$

No prime is being assigned a distinct tail denominator: the component bound explicitly accounts for shared denominators.

Define, for $0 < x \leq e$,

$$w(x) = \sum_{\substack{m \geq 1 \\ 1 \leq mx \leq e}} \left(\frac{1}{mx} - \frac{1}{e} \right). \quad (14)$$

On any fixed interval $[\varepsilon, e]$, this function has finitely many pieces and is bounded and Riemann integrable. Away from finitely many endpoints, $NW_N(p)$ tends uniformly to $w(p/N)$; moving endpoints arising from $L/N - e = o(1)$ contribute $o(N/\log N)$ to the relevant prime sums. This last statement follows by enclosing the finitely many endpoints in fixed intervals of total length at most an arbitrary $\rho > 0$, using the prime number theorem there, then letting $\rho \downarrow 0$.

The prime number theorem and (13) therefore give

$$\liminf \frac{D \log N}{N} \geq e \int_{\varepsilon}^e \min \left\{ w(x), \frac{1}{h} \left(\frac{1}{e} - \frac{1}{T} \right) \right\} dx.$$

Let $T \rightarrow \infty$, then $\varepsilon \downarrow 0$. The integrand is bounded by $1/(eh)$, so these limits are valid even though $w(x)$ itself grows near zero. This proves (6) with

$$\mathcal{C}(h) = e \int_0^e \min \left\{ w(x), \frac{1}{eh} \right\} dx. \quad (15)$$

6. Proof of Theorem 3

Again reduce to $D = O(N/\log N)$ and fix ε, T . First assume $M \leq N^{B+o(1)}$, $1 < B < 2$, and let $s = \lceil 1/(B-1) \rceil$. Every denominator contains at most one prime greater than εN , and that prime occurs to the first power, for sufficiently large N .

For a prime $p \in V$, write the sum of all selected low multiples of p as $U/(pV_0)$, where U, V_0 are positive integers bounded uniformly in terms of ε, T . Write all its tail multiples as $px_1, \dots, px_r$. If $r \leq s-1$, then

$$x_i \leq \frac{M}{\varepsilon N} \leq N^{B-1+o(1)}.$$

The numerator after combining these terms is the positive integer

$$Z = U \prod_{i=1}^r x_i + V_0 \sum_{i=1}^r \prod_{j \neq i} x_j.$$

The denominator is $pV_0 \prod x_i$, with every factor other than p coprime to p . The total reciprocal sum being 1 therefore requires $p \mid Z$. But

$$0 < Z \leq N^{r(B-1)+o(1)} = o(N) < p,$$

since $(s-1)(B-1) < 1$. For $r = 0$, interpret $Z = U$; the same contradiction applies. Thus each prime in V has at least s tail multiples. Distinct such primes cannot share a denominator, so $r_T \geq s|V|$.

Use this stronger count in (13). The prime sum now has limiting integrand $\min\{w(x), s(1/e - 1/T)\}$. The same limit argument proves (7a).

For (7b), the assumption also implies $M = o(N^2)$, so the same disjointness and prime-power observations apply. For $r \leq s-1$,

$$\left(\frac{M}{\varepsilon N}\right)^r = o(N).$$

Consequently $Z = o(N)$ again, including the case $r = s-1$.

Finally, integral comparison in (14) gives

$$w(x) = \frac{1}{ex} + O(1) \quad (x \downarrow 0).$$

Since taking a minimum with a fixed constant is a Lipschitz operation, comparison of the integrands with $\min(1/x, s)$ on $(0, e)$ gives

$$d_s = \int_0^e \min(1/x, s) dx + O(1) = \log s + O(1).$$

7. Evaluation of the constants

First, $w(x) \geq 1/e$ for $0 < x \leq 1/2$. To see this, w decreases on each interval $(1/(m+1), 1/m)$: when a term disappears at $x = e/j$, its value is zero. Its infimum there is the limit at $1/m$ from below. For $m = 2$, this limit is

$$2 \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right) - \frac{3}{e} = \frac{47}{30} - \frac{3}{e} > \frac{1}{e}.$$

For $m \geq 3$, comparison with an integral gives the lower bound

$$g(m) := \int_{m+1}^{em} \left(\frac{m}{u} - \frac{1}{e} \right) du = \frac{m+1}{e} - m \log \left(1 + \frac{1}{m} \right).$$

The function $g(y)$ is increasing for $y \geq 3$, since

$$g'(y) = \frac{1}{e} - \log(1 + 1/y) + \frac{1}{y+1} > \frac{1}{e} - \frac{1}{y(y+1)} > 0.$$

Also $g(3) > 1/e$, because $e \log(4/3) < 1$. Values at the jump points themselves are larger, so the stated bound follows.

On $[1/2, 1)$, w is decreasing. Its limit at 1 from below is $1/2 - 1/e$, which exceeds $1/(3e)$, since $e > 8/3$. The relevant crossings are:

- $w(x) = 1/e$ at $x = 5e/18$; between this point and $e/3$, $w(x) = 5/(6x) - 2/e$.
- $w(x) = 1/(2e)$ at $x = e/3$; between $e/3$ and 1, $w(x) = 1/(2x) - 1/e$.
- For $t \geq 3$, there is no crossing below 1.

On $[1, e]$, the crossing with $1/(et)$ occurs at $x = et/(t+1)$. Below it the minimum in (15) is $1/(et)$; above it $w(x) = 1/x - 1/e$. Thus

$$e \int_1^e \min\{w(x), 1/(et)\} dx = e \log(1 + 1/t) - \frac{1}{t}. \quad (16)$$

For $t \geq 3$, the interval $(0, 1)$ contributes exactly $1/t$. The same crossing and saturation calculation holds for every real $h \geq 3$, giving $\mathcal{C}(h) = e \log(1 + 1/h)$ and hence the asserted formula for every real $B \geq 4$. For $t = 1, 2$, integrating the two explicitly displayed formulas below 1 gives (7). The decimal values are only illustrations; the exact formulas and their proof establish the bounds.

For the headline subquadratic constant, exact integration gives

$$d_2 = e \left[\frac{19}{20} \log \frac{20}{19} + \frac{77}{60} \log 3 + \frac{13}{12} \log \frac{15}{13} + \frac{7}{3} \log \frac{4}{3} + \log 2 \right] - \frac{11}{2}.$$

The regions where $w(x) < 2/e$, apart from endpoints, are

$$\left(\frac{19e}{120}, \frac{1}{2} \right), \quad \left(\frac{13e}{60}, 1 \right), \quad \left(\frac{3e}{8}, e \right).$$

Split the first at $e/6$, the second at $e/4, e/3$, and the third at $e/2$. The seven formulas for $w(x) = A/x - n/e$ have respective pairs

$$(A, n) = (19/20, 4), (47/60, 3), (13/12, 3), (5/6, 2), (1/2, 1), (3/2, 2), (1, 1).$$

Integrating these pieces gives the boxed expression. To verify that no interval near zero is omitted, $w(x) > 2/e$ for $x \leq 1/3$: at the left limit of $1/3$, use $3 \sum_{j=4}^8 1/j = 743/280$ and $e > 8/3$; for the subsequent left limits use $g(m) \geq g(4) > 2/e$. This last inequality follows from $\log(5/4) < 1/4$ and $e < 3$. Monotonicity between jumps then supplies exactly the listed crossings.

8. Proof of Theorem 4: comparison across tail scales

Fix $C \geq 0$, and write $q = \lfloor C + b \rfloor$. Corollary 1.1 settles the case $q = 0$, so assume $q \geq 1$. All estimates in this section are uniform over the representations in question once C is fixed. We work first with starts in $[X, 2X]$, for large X .

8.1. The core and a bounded exponential range for the tails

For each $N \in \mathcal{E}_C \cap [X, 2X]$, choose a minimum-length representation A_N . Split it into the core $B_N = A_N \cap [N, N^2]$ and r_N tails above N^2 . Theorem 1 gives $1 \leq r_N \leq q$. Put

$$L_N = N + |B_N| - 1, \quad I_N = [N, L_N] \cap \mathbb{Z}.$$

Harmonic comparison and $r_N = O_C(1)$ imply

$$L_N = eN + O_C(1), \quad 0 \leq \Delta_N := \sum_{n \in I_N} \frac{1}{n} - \sum_{a \in B_N} \frac{1}{a} \leq \frac{C+2}{N}. \quad (\text{D1})$$

For clarity, the upper bound follows from the total comparison interval $[N, N + |A_N| - 1]$: shortening it by r_N removes more reciprocal mass than deleting the actual tails above N^2 . Its excess harmonic mass is at most $(C+2)/N$ by (9). The lower bound is harmonic extremality at the core's cardinality.

Let $H_N = I_N \setminus B_N$, $F_N = B_N \setminus I_N$, and $t_N = |H_N| = |F_N|$. The loss identity gives

$$\Delta_N = \sum_{h \in H_N} \left(\frac{1}{h} - \frac{1}{L_N} \right) + \sum_{f \in F_N} \left(\frac{1}{L_N} - \frac{1}{f} \right) \geq \frac{t_N(t_N - 1)}{2L_N^2}.$$

Thus $t_N = O_C(\sqrt{N})$. The reduced denominator v_N of the positive residual of B_N divides $\text{lcm}(1, \dots, L_N) \prod_{f \in F_N} f$. Since $f \leq N^2$, the prime number theorem yields

$$\log v_N \leq (e + o_C(1))N. \quad (\text{D2})$$

We also need an elementary upper bound for a fixed number of unit fractions. If $u/v = \sum_{i=1}^r 1/x_i > 0$, with $x_1 < \dots < x_r$, set $T_j = v \prod_{i=1}^j x_i$. The remaining positive residual has denominator dividing T_j , so

$$\frac{1}{T_j} \leq \sum_{i=j+1}^r \frac{1}{x_i} \leq \frac{r-j}{x_{j+1}}, \quad T_{j+1} \leq (r-j)T_j^2.$$

Iteration, followed by $x_r \leq T_{r-1}$, proves

$$x_r \leq K_r v^{2^{r-1}}, \quad K_1 = 1, \quad K_r = \prod_{t=2}^r t^{2^{t-2}}. \quad (\text{D3})$$

Together, Theorem 1 and (D2)–(D3) give the convenient coarse bounds

$$\frac{X}{q} \leq \max_{a \in A_N} \log a \leq 2^{q+3} X. \quad (\text{D4})$$

For two starts separated by $d = |N' - N| = o(X)$, the comparison intervals overlap. Their endpoints differ by at most $ed + O_C(1)$, whence

$$|B_N \triangle B_{N'}| \leq (e+1)d + O_C(\sqrt{X}), \quad \max(B_N \cup B_{N'}) \leq 4X^2. \quad (\text{D5})$$

8.2. A comparison that preserves the lower tails

Fix $17/30 < \theta < 0.99$, put $H_0 = X^\theta$, and set

$$R = 100(q+1), \quad A_0 = 32(q+1), \quad \mathcal{W}(h) = \frac{h}{100(e+1) \log(4X^2)}.$$

From [8, Corollary 1.3], for all sufficiently large X ,

$$\#\{p \text{ prime} : n \leq p < n + d\} \geq \frac{d}{8 \log X} \quad (\text{D6})$$

whenever $[n, n + d] \subseteq [X, 2X]$ and $d \geq 4H_0$. Indeed, use at least $d/(2H_0)$ consecutive intervals of length H_0 lying before the right endpoint. Each contains at least $H_0/(2 \log(2X))$ primes, after harmless endpoint adjustments. Choose a fixed $0 < \eta < \theta - 17/30$ to meet the source's lower length threshold. Its upper threshold also holds because $\theta < 0.99$.

Consider any family in which the tails at each start are partitioned as Y_N, G_N, Z_N , with at most q terms in total. Suppose the reciprocal sum of Z_N is constant throughout the family, G_N is nonempty with every $\log a \geq h$, and every tail in Y_N has $\log a < \ell$, where

$$H_0 \leq \ell \leq h/R, \quad h \leq 2^{q+3}X.$$

Comparison claim. Within any interval of starting values of length at most $\mathcal{W}(h)$, the reciprocal sum of G_N is constant, and the occupied starts have diameter less than $A_0\ell$.

For two such starts, cancellation of the full sums and of the sums of Z_N gives

$$\sum_{a \in G_N} \frac{1}{a} - \sum_{a \in G_{N'}} \frac{1}{a} = - \left(\sum_{a \in B_N} \frac{1}{a} - \sum_{a \in B_{N'}} \frac{1}{a} \right) - \left(\sum_{a \in Y_N} \frac{1}{a} - \sum_{a \in Y_{N'}} \frac{1}{a} \right). \quad (\text{D7})$$

A denominator for the right side has logarithm at most

$$((e+1)d + O_C(\sqrt{X})) \log(4X^2) + 2q\ell \leq h/100 + o_{C,\theta}(h) + 2qh/R < h/2.$$

Here $\theta > 1/2$ controls the edits uniformly for $h \geq H_0$, and the bound on h ensures $d = o_C(X)$. The left side of (D7) has absolute value at most $2qe^{-h}$. This is smaller than the absolute value of any nonzero rational with the allowed denominator. It must therefore vanish. Equality of the individual tails is not needed.

It follows that the reciprocal sums of $B_N \cup Y_N$ agree. For $N < N'$ in this interval, eventually

$$N' \leq 21N/20, \quad 5N/2 \leq L_N, L_{N'} \leq 29N/10.$$

For each prime $p \in [N, N')$, the old comparison block contains $p, 2p$, the new one contains $2p$, and neither contains $3p$. Unless an edited core position or a denominator in $Y_N \cup Y_{N'}$ is divisible by p , the equal-sum identity has an uncanceled $1/p$ and is impossible modulo p .

Each relevant edited occurrence costs at least $1/(87N)$ in its own loss identity. A hole is at most $2p \leq 21N/10$, with cost at least $10/(21N) - 2/(5N) = 8/(105N)$. An addition is at least $3p \geq 3N$, with cost at least $10/(29N) - 1/(3N) = 1/(87N)$. The two cores' total loss is at most $2(C+2)/N$. Each edited position is at most $(N')^2 < N^3$, so contains at most two distinct primes at least N . Thus edits witness at most $348(C+2)$ primes. The distinct remaining primes divide the product of the lower tails, whose logarithm is less than $2q\ell$. Hence

$$\#\{p \text{ prime} : N \leq p < N'\} \leq 348(C+2) + \frac{2q\ell}{\log X}. \quad (\text{D8})$$

If $d \geq A_0\ell$, (D6) would give a lower bound $4(q+1)\ell/\log X$, contradicting (D8) for large X . This proves the comparison claim, including the lower-tail contribution.

8.3. Group the heights, then retain each localization

Use the grid $H_j = R^j H_0$, $j \geq 0$. A tail with $\log a < H_0$ belongs to the base bin, labelled -1 ; all others belong to bin j when $H_j \leq \log a < H_{j+1}$. By (D4), only $O_C(\log X)$ labels occur. Classify a start by the multiset of labels of all its tails. There are $O_C((\log X)^q)$ possible classes.

Fix a class. Adjoin the label -1 to its occupied labels even if the base bin has no actual tails. Group consecutive occupied indices into maximal components. The component containing -1 is the bottom group. It includes bin 0 whenever that bin is occupied; there is then no height gap allowing those tails to be separated from the base.

Process every other component from highest to lowest. For a component $[u, v]$, let $h = H_u$, $U = H_{v+1}$. It has at most q occupied bins, so

$$h \leq \log a < U \leq R^q h. \quad (\text{D9})$$

Let ℓ be the upper endpoint of the next lower component, or of the bottom group if it comes next. The missing bin gives $H_0 \leq \ell \leq h/R$. All unprocessed lower tails have height less than ℓ . The bottom group's upper endpoint is at most $R^q H_0$.

There is a nonbottom component for large X , since otherwise the last bound contradicts (D4). At the top component, $h_{\text{top}} \geq X/(qR^q)$. Partition $[X, 2X]$ into $O_C(\log X)$ intervals of length at most $\mathcal{W}(h_{\text{top}})$. Apply the comparison claim on each occupied interval, with Z_N empty. The top group's sum becomes constant there, and its occupied starts now lie within a diameter less than $A_0 \ell$.

Retain this smaller interval. If another component remains, its upper height endpoint is this ℓ . By (D9), the parent's diameter is at most $A_0 R^q h_{\text{next}}$. Dividing it into intervals of width $\mathcal{W}(h_{\text{next}})$ therefore requires only $O_C(\log X)$ children. On the parent, the sum of all already processed groups is constant. Apply the comparison claim in each child to fix the current group's sum and obtain the next, smaller diameter.

There are at most q processed components. Each class consequently produces at most $O_C((\log X)^q)$ terminal families, each of diameter $O_C(H_0)$, because only the bottom group remains. Constants fixed in different families are never identified. Every unprocessed group is included in (D8) until its sum has been fixed.

Multiplying the number of classes, the number of terminal families per class, and the number of integers in each terminal interval gives

$$\#(\mathcal{E}_C \cap [X, 2X]) \ll_{C,\theta} X^\theta (\log X)^{2q}.$$

Sum dyadic intervals. For a prescribed $\varepsilon > 0$, choose $17/30 < \theta < \min(17/30 + \varepsilon, 0.99)$ and absorb the fixed logarithmic power into the remaining power of X . This proves (D). It bounds the number of exceptional starts, without proving their eventual absence. $\square$

9. Proof of Theorem 5: constructions under quadratic ceilings

The lower bounds in (Q)–(Q1) follow from Theorem 2 with $B = 2$. We prove the upper bounds. The cancellation mechanism is classical; see Croot and Martin [2,3], particularly [3, Lemmas 14–15].

9.1. A scaled two-fraction device and a binary finish

For an odd prime power $q = p^a \geq 5$ and any residue $t \pmod{p}$, there are distinct integers m_1, m_2 , coprime to p , with

$$(q-3)/2 \leq m_1 < m_2 < q, \quad m_1^{-1} + m_2^{-1} \equiv t \pmod{p}. \quad (\text{Q2})$$

Here and below inverses in congruences are modular inverses. For $p \geq 5$, the $(p+3)/2$ integers immediately below q are nonzero and distinct modulo p . Their inverses form a set X of that size. Since $|X \cap (t-X)| \geq 3$, and at most one element gives an equal pair, an unequal pair exists. Its members are at least $q-(p+3)/2 \geq (q-3)/2$. For $p = 3$, necessarily $q \geq 9$; the pairs $(q-2, q-1)$, $(q-4, q-1)$, $(q-5, q-2)$ give inverse sums 0, 1, 2, respectively. This proves (Q2). In particular, $m_i \geq q/5$, and every odd prime power dividing m_i is smaller than q .

Suppose $S = u/v$ is a reduced running sum, and its largest odd prime-power divisor is $q = p^a \leq N$. Write $v = qv_0$. For $q \geq 5$, choose a power of two Q so that

$$N^2/2 \leq Qq^2 \leq N^2.$$

Take $Q = 1$ if $N^2/(2q^2) \leq 1$, and otherwise the least power of two at least that ratio. Apply (Q2) with $t \equiv -Quv_0^{-1} \pmod{p}$, and add $1/(Qqm_1) + 1/(Qqm_2)$. Multiplication of the new sum by q gives residue zero modulo p , so the exponent of p in its reduced denominator drops. Every other odd prime power introduced is smaller than q . The largest remaining odd prime power decreases strictly, and each new denominator is in

$$[N^2/10, N^2]. \quad (\text{Q3})$$

If $q = 3$, take Q as the largest power of two at most $N^2/6$. The nonzero target $-Q(3S) \pmod{3}$ equals a^{-1} for an $a \in \{1, 2\}$. Adding $1/(3Qa)$ removes 3, with denominator in $(N^2/4, N^2]$.

If all initial odd prime powers are at most $Y \leq N$, this procedure uses at most $2\pi^*(Y)$ fractions, where $\pi^*(Y)$ counts all prime powers up to Y . A stage's new denominators have largest odd prime-power divisor exactly its label q . Thus stages give disjoint denominators, and the two within a stage are distinct. We use

$$\pi^*(Y) = \pi(Y) + O(\sqrt{Y} \log Y).$$

For $Y = N$ or $Y = \varepsilon N$ with fixed $\varepsilon > 0$, the total added reciprocal mass is $O_\varepsilon(1/(N \log N))$.

In both constructions below, the initial core lies in $[N, L]$, with $L = eN + O(N/\log N) > 2N$. Remove its largest power of two d , and arrange that the resulting residual satisfies

$$1/d < R_0 \leq 1/d + 1/(L+1) < 2/d. \quad (\text{Q4})$$

In particular $R_0 > 1/L$. The corrections have total mass $o(1/N)$, so the final residual is positive, dyadic, and less than $2/d$. Its finite binary expansion has distinct denominators at least $d \geq N$. These miss the core, because d was removed and every larger power of two exceeds L , and miss the corrections, which have odd prime factors. When the maximum two-adic prime power in all denominators is at most N^2 , the binary expansion also has denominators at most N^2 and uses $O(\log N)$ terms.

9.2. The upper bound at the exact ceiling N^2

Call an integer eligible if every odd prime power dividing it is at most N . For $X = eN + O(N/\log N) < 3N$, the excluded integers caused by primes greater than N are precisely p for $N < p \leq X$, and $2p$ for $N < p \leq X/2$. Exclusions caused by nonprime odd prime powers greater than N number $O(\sqrt{N} \log N)$,

each such power having at most two multiples below $3N$. They contribute $o(1/\log N)$ harmonic mass. Put

$$\mu_0 = \frac{3}{2} - \frac{1}{2} \log 2, \quad \nu_0 = \frac{3e}{2} - 2.$$

By the prime number theorem and partial summation, for each fixed real c ,

$$\sum_{\substack{N \leq n \leq eN + cN/\log N \\ n \text{ eligible}}} \frac{1}{n} = 1 + \frac{c/e - \mu_0 + o(1)}{\log N}. \quad (\text{Q5})$$

The removed prime mass is $\sum_{N < p \leq eN} 1/p + \frac{1}{2} \sum_{N < p \leq eN/2} 1/p = (\mu_0 + o(1))/\log N$; the removed count has coefficient ν_0 . Moving the endpoint by $O(N/\log N)$ changes those removed contributions only by lower-order terms, as follows from prime number theorem sums on fixed scaled intervals and then shrinking neighborhoods of their endpoints.

Let $L+1$ be the first denominator at which the eligible reciprocal sum reaches or exceeds one. Equation (Q5), bracketed by $c = e\mu_0 \pm \eta$, proves its existence and

$$L = eN + (e\mu_0 + o(1))N/\log N.$$

Take the eligible core through L , removing its largest power of two d . The definition of the first crossing gives (Q4) without any assumption about gaps between eligible integers. Its count is

$$k_0 = (e-1)N + \left(2 - \frac{e}{2} \log 2 + o(1)\right) \frac{N}{\log N}. \quad (\text{Q6})$$

Apply Section 9.1 with $Y = N$. There are at most $(2 + o(1))N/\log N$ added terms, all above L for large N . The mass bound leaves a positive residual, and every initial or added denominator is at most N^2 . Hence the binary finish also respects that bound. The distinctness checks in Section 9.1 cover all terms. Equation (Q6) proves the upper bound in (Q1).

9.3. An optimized core for the ceiling $(e^2/6)N^2$

Put

$$\alpha = 5e/18, \quad \beta = e/2, \quad \kappa = e^2/6, \quad \mathcal{O} = (\alpha, 1) \cup (\beta, e], \quad \mathcal{K} = (0, \alpha] \cup [1, \beta].$$

Fix $0 < \varepsilon < 1/10$. All implicit constants in the remainder of the construction may depend on this fixed cutoff. An integer is eligible if it has neither a prime divisor with $p/N \in (\alpha, 1) \cup (\beta, \infty)$, nor a nonprime odd prime-power divisor exceeding εN . Pure powers of two are eligible. Higher-power exclusions up to $3N$ number $O_\varepsilon(\sqrt{N} \log N)$; each candidate power exceeds εN , so has only $O_\varepsilon(1)$ multiples there. Distinct omitted primes cannot share a multiple below $3N$, since both exceed αN .

Define

$$g(x) = \sum_{1 \leq mx \leq e} \frac{1}{mx}, \quad f(x) = \#\{m \geq 1 : 1 \leq mx \leq e\}, \quad w(x) = g(x) - f(x)/e,$$

and put $\mu = \int_{\mathcal{O}} g(x) dx$, $\nu = \int_{\mathcal{O}} f(x) dx$. Prime number theorem sums give

$$\sum_{\substack{N \leq n \leq eN + cN/\log N \\ n \text{ eligible}}} \frac{1}{n} = 1 + \frac{c/e - \mu + o_\varepsilon(1)}{\log N}. \quad (\text{Q7})$$

Only finitely many multiplier intervals occur on $\mathcal{O}$, so the same fixed-endpoint argument as for (Q5) handles the moving endpoint. Define $L + 1$ by the first crossing, and remove the largest power of two $d \leq L$. Then (Q4) holds, and

$$L = eN + (e\mu + o_\varepsilon(1))N/\log N > eN, \quad k_0 = (e - 1)N + \left(e \int_{\mathcal{O}} w(x) dx + o_\varepsilon(1)\right) \frac{N}{\log N}. \quad (\text{Q8})$$

For later evaluation, direct integration yields

$$\mu = \frac{5}{6} \log \frac{6}{5} + \frac{1}{2} \log \frac{3}{e} + \log 2, \quad \nu = 1 + \alpha. \quad (\text{Q9})$$

Here $f = 2$ on $(\alpha, e/3)$, and $f = 1$ on each of $(e/3, 1)$ and (β, e) . In particular $\mu > 0$, justifying the positive endpoint displacement in (Q8).

Every retained prime $p > \varepsilon N$ lies in $(\varepsilon N, \alpha N] \cup [N, \beta N]$, and all its multiples in $[N, L]$ belong to the core. Their cofactors are at most $3/\varepsilon$, so cannot contain an omitted prime or a forbidden prime power for large N . Such a multiple is not the removed power of two. Write their reciprocal sum as

$$\frac{1}{p} \frac{U}{V}, \quad \frac{U}{V} = \sum_{N \leq mp \leq L} \frac{1}{m}, \quad (U, V) = 1. \quad (\text{Q10})$$

The positive integers U, V lie in a finite set depending on ε , and eventually $p \nmid UV$. Every correction below has cofactors whose odd prime powers are at most εN . Consequently a correction at one high prime leaves all other coefficients (Q10) unchanged. For $N \leq p \leq \beta N$, (Q8) ensures $2p \leq eN < L < 3p$, so $U/V = 3/2$ on the entire upper interval.

9.4. One fraction for almost all retained high primes

Let $1 \leq a_p < p$ be the unique integer with $a_p \equiv -VU^{-1} \pmod{p}$. Then

$$Ua_p + V = jp, \quad a_p = (jp - V)/U, \quad 1 \leq j \leq U \quad (\text{Q11})$$

for sufficiently large N . In particular $a_p \geq p/(2U) \gg_\varepsilon N$. If every odd prime power in a_p is at most εN , call p good and add $1/(pa_p)$. Its coefficient cancels p modulo p , with no other large odd prime power introduced.

There are $o_\varepsilon(N/\log N)$ bad primes. First, integers $a \leq \beta N$ with a nonprime odd prime-power divisor greater than εN number $O_\varepsilon(\sqrt{N} \log N)$. For each fixed (U, V, j) , (Q11) determines p uniquely from a_p . There are only finitely many such tuples. Second, if $a_p = dq$ with a prime $q > \varepsilon N$, then $1 \leq d \leq \beta/\varepsilon$ and

$$jp - Udq = V. \quad (\text{Q12})$$

For a fixed tuple (U, V, j, d) , the upper sieve for two linear forms [9, Theorem 2.5] bounds the number of prime pairs in (Q12) by $O_\varepsilon(N/\log^2 N)$. To check its hypotheses, divide by $\gcd(j, Ud)$ when that gcd divides V ; otherwise there are no solutions. Parametrizing the integer solutions gives two linear forms with nonzero determinant, because the fixed shift V is positive. Nonprimitive forms or a local obstruction give at most finitely many prime solutions. In the remaining case the forms are primitive, distinct, and admissible, so the cited upper sieve applies. All coefficients are fixed once the tuple and ε are fixed. Thus

$$\#\{p \text{ bad}\} = O_\varepsilon(N/\log^2 N) + O_\varepsilon(\sqrt{N} \log N) = o_\varepsilon(N/\log N). \quad (\text{Q13})$$

For every bad prime use two fractions. Among integers $p/12 \leq m < 2p/3$, those with all odd prime powers at most εN number $(7/12 + o_\varepsilon(1))p$, uniformly for $\varepsilon N < p \leq \beta N$. Indeed, prime divisors above

εN remove only $O_\varepsilon(N/\log N)$ integers, their complementary factors being bounded; higher powers remove $O_\varepsilon(\sqrt{N} \log N)$ more. Both are $o_\varepsilon(p)$.

All these m are nonzero modulo p , and their inverses form a set X_p with more than $p/2$ elements. More precisely,

$$|X_p \cap (-UV^{-1} - X_p)| \geq 2|X_p| - p = (1/6 + o_\varepsilon(1))p.$$

At most one element represents an equal pair. Hence distinct allowed m_1, m_2 satisfy $m_1^{-1} + m_2^{-1} \equiv -UV^{-1} \pmod{p}$. Adding $1/(pm_1) + 1/(pm_2)$ cancels p , again introducing no other odd prime power above εN .

9.5. Cap, mass, distinctness, and the constant

For $p \leq \alpha N$, good denominators are less than $p^2 \leq \alpha^2 N^2 < N^2$; bad-pair denominators are smaller still. For $N \leq p \leq \beta N$, the coefficient is $3/2$, so $a_p = (jp - 2)/3$ in (Q11). The possibility $j = 3$ is excluded modulo 3. Thus $j \in \{1, 2\}$, and every good denominator obeys

$$pa_p < \frac{2}{3}p^2 \leq \frac{2}{3}\beta^2 N^2 = \kappa N^2. \quad (\text{Q14})$$

The bad-pair cofactors also satisfy $m_i < 2p/3$, giving the same exact cap. Every high-prime correction is at least $c_\varepsilon N^2$, by (Q11) or $m_i \geq p/12$. They therefore exceed L , and their total mass is $O_\varepsilon(1/(N \log N))$.

After all high primes are removed, every remaining odd prime power is at most εN . Apply Section 9.1 with $Y = \varepsilon N$. It adds at most

$$(2\varepsilon + o_\varepsilon(1))N/\log N \quad (\text{Q15})$$

fractions, all in $[N^2/10, N^2]$, with total mass $o_\varepsilon(1/N)$. Equation (Q4) therefore leaves a positive dyadic residual. The largest two-adic prime power is at most N^2 : for a high-prime correction it divides its cofactor, which is smaller than p , and for a small-power correction its full denominator is at most N^2 . The binary finish meets the cap and uses $O(\log N)$ terms.

All denominators are distinct. High-prime stages have different largest odd prime-power divisors $p > \varepsilon N$. Within a bad stage the two cofactors differ. The later small-power stages have strictly decreasing largest odd prime powers at most εN . All corrections exceed the core endpoint, and none is a power of two. The reserved power of two supplies the remaining binary distinctness.

The number of retained high primes is

$$(\alpha + \beta - 1 - \varepsilon + o_\varepsilon(1))N/\log N.$$

All but the negligible set (Q13) cost one fraction each. Combining (Q8), (Q15), and the binary count, the excess is at most

$$\left(e \int_{\mathcal{O}} w(x) dx + \alpha + \beta - 1 + \varepsilon + o_\varepsilon(1) \right) N/\log N. \quad (\text{Q16})$$

Section 7's crossings show that $w(x) < 1/e$ on $\mathcal{O}$ and $w(x) \geq 1/e$ on $\mathcal{K}$, apart from endpoints. Since $|\mathcal{K}| = \alpha + \beta - 1$,

$$e \int_{\mathcal{O}} w(x) dx + \alpha + \beta - 1 = e \int_0^e \min\{w(x), 1/e\} dx = c_1.$$

The same identity follows directly from (Q9). For each fixed ε , (Q16) proves the eventual upper bound under the same exact ceiling κN^2 . Take $N \rightarrow \infty$ first, then $\varepsilon \downarrow 0$. This proves (Q). It uses no uniform sieve estimate for varying ε . $\square$

10. Scope and remaining gap

Theorem 1 implies that fixed additive excess requires exponential denominators, and Theorems 2 and 3 give growing lower bounds under polynomial denominator ceilings. Theorem 5 supplies upper constructions and a matching leading constant at its stated ceiling. These restrictions do not exclude boundedly many exponential denominators in the unrestricted problem.

Theorem 4 shows that, for each fixed excess bound, exceptional starts form a set of density zero with a quantitative counting bound. It still permits infinitely many exceptional starts. No argument here excludes bounded excess at every sufficiently large starting value, so the pointwise limit in Erdős problem 295 remains unproved. No priority claim is made.

AI assistance and validation. OpenAI GPT-6 through Codex contributed substantially to the derivation, writing, informal proof reviews, and literature searches. The revision was checked locally against the hypotheses of its cited inputs, and selected finite arithmetic and grouping steps were tested exactly. Such tests do not establish the asymptotic theorems; those rely on the written proofs. The additions have not received independent peer review and are not proof-assistant formalizations.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique 28 (1980), pp. 34-35. https://mathweb.ucsd.edu/~ronspubs/80_11_number_theory.pdf
- [2] E. S. Croot III, *On unit fractions with denominators in short intervals*, Acta Arithmetica 99 (2001), 99-114. In particular, the prime-divisor discussion on p. 105 and Lemma 4. <https://arxiv.org/abs/math/9904181>
- [3] G. Martin, *Denser Egyptian fractions*, Acta Arithmetica 95 (2000), 231-260. See Section 5 for related prime-power cancellation methods. <https://matwbn.icm.edu.pl/ksiazki/aa/aa95/aa9533.pdf>
- [4] M. D. Vose, *Egyptian fractions*, Bulletin of the London Mathematical Society 17 (1985), 21-24. Original full proof not inspected; the uniform theorem was checked through [5]. <https://doi.org/10.1112/blms/17.1.21>
- [5] G. Tenenbaum and H. Yokota, *Length and denominators of Egyptian fractions, III*, Journal of Number Theory 35 (1990), 150-156. <https://tenenb.perso.math.cnrs.fr/PPP/Egyptian.pdf>
- [6] H. D. Ruderman, P. Erdős and E. G. Straus, *E2232*, American Mathematical Monthly 78 (1971), 302-303. Original problem and solution inspected for this revision. <https://doi.org/10.2307/2317539>
- [7] N. Burshtein, *On distinct unit fractions whose sum equals 1*, Discrete Mathematics 300 (2005), 213-217. Bibliographic reference; full text not inspected. <https://doi.org/10.1016/j.disc.2004.11.007>
- [8] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, arXiv:2405.20552v2 (7 April 2026), Corollary 1.3, pp. 2-3. <https://arxiv.org/abs/2405.20552v2>
- [9] K. Ford, *Sieve Methods Lecture Notes*, Spring 2023, Theorem 2.5, pp. 19-21. <https://ford126.web.illinois.edu/sieve2023.pdf>